Technical Challenges and Risk Mitigation

Convertible Wheelchair-to-Bed Transfer System

Problems found while designing, and how they are fixed

Problem	Modification	Result
The sliding boards were too thick: At 41 mm thick, the boards would either press into the mattress or drop the patient by about 4 cm as they came off.	Smaller rollers and thinner boards are used	The boards are now 28 mm thick and glide about 3 mm above the mattress. The patient will barely notice the step.
Reclining the chair was too heavy: Lowering the backrest with a patient on it needed about 214 N, which is like handling a 22 kg mass.	The gas spring is repositioned so it helps most where the load is highest, and the grip is moved to the head end for better leverage.	Reclining now takes at most about 76 N, whether the chair is empty or holding a 120 kg patient.
The crank was hard to turn at the lowest height: At its lowest position, raising the chair needed about 139 N on the crank handle.	The crank handle is made longer, so the same effort turns the screw more easily.	The hardest point is now about 115 N, and only for the first few turns. Above 55 cm, it is under 80 N.
The backrest frame was not strong enough: With the chair flat and someone leaning on the head end, the side tubes of the backrest were working closer to their limit than is comfortable.	The tube walls are made thicker (2 mm to 3 mm).	The backrest frame now has a comfortable safety margin of more than 3 times its working load.
The docking arm went through the bed frame, and fitted only one bed: The first docking arm simply ran into the bed frame. A later version could grip only one particular bed, so thinner or thicker mattresses or a gap between chair and bed would stop it from working.	The clamp is redesigned: an arm that swings out to reach the bed, and two jaws that slide up and down in steps and lock with a pin, finished off with two hand-tightened screws.	One clamp design now fits mattresses from 9.5 to 28 cm thick, bed frames 2 to 10.5 cm deep, and a gap of up to 8 cm between chair and bed.

Potential risks while handling the chair, and their solutions

Problem	Solution
The backrest drops suddenly: The patient falls back with the backrest.	Two separate locks: the gas spring locks itself whenever the lever is let go, and a spring pin clicks into place when the back is fully upright or fully flat. Releasing it takes two separate actions.
The patient rolls towards the caregiver: The patient could fall on the far side.	The transfer bar works as a side rail, and the armrest on the caregiver's side stays on.
The sliding boards rub or press on the skin: This could cause discomfort or skin damage.	The boards glide 3 mm above the mattress. When the boards are pulled out, the belt rolls freely (in free mode) so it does not drag on the skin.
Gaps between the seat sections under the hips or knees: Pressure or pinching under the patient.	Smooth filler strips and covers go over the gaps, and the hips rest on the seat section, not on a gap.
The chair rolls away or tips during a transfer: The patient falls between chair and bed.	All four wheels lock with a pedal. The chair is clamped to the bed frame, and a hand-tight clamp holds about 8 times more than needed.
A sliding board comes out while the chair is moving: The patient is shifted unexpectedly.	Each board is held by a lock pin that is locked by default. Pull and quarter-turn to release, and it snaps back in by itself when the board slides home.
The lift slowly sinks: The chair drifts down under the patient.	The lifting screw cannot be turned back by the patient's weight. An extra friction washer and end stops add a second safeguard.
The wrong belt mode is chosen: The patient moves when they should not, or the board drags.	Push/pull handles show the mode at a glance, and checking the mode is part of the checklist.